

AN ANALYSIS OF THE LARGEST 250 PORTUGUESE COMPANIES

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OBJECTIVE

Understand the composition of the largest portuguese companies and find meaningful patterns in their economic and financial structure.

DATA

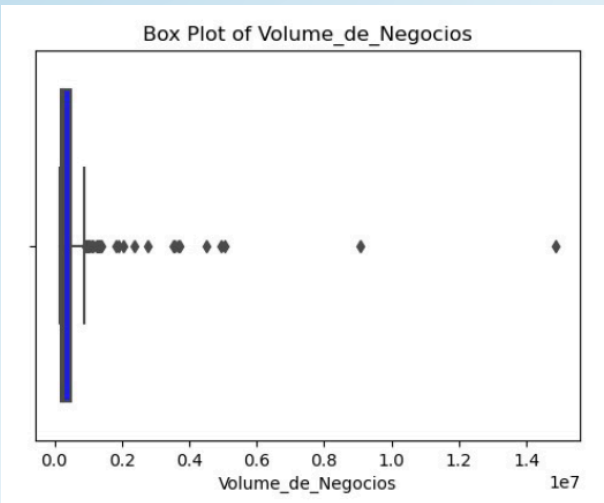
- Top 250 Portuguese companies by Turnover (Total Sales of Goods and Services) in 2022. Source: SABI database

19 variables representing economic, financial and activity attributes of the companies

For the Cluster Analysis:

10 meaningful economic and financial ratios

UNIVARIATE ANALYSIS

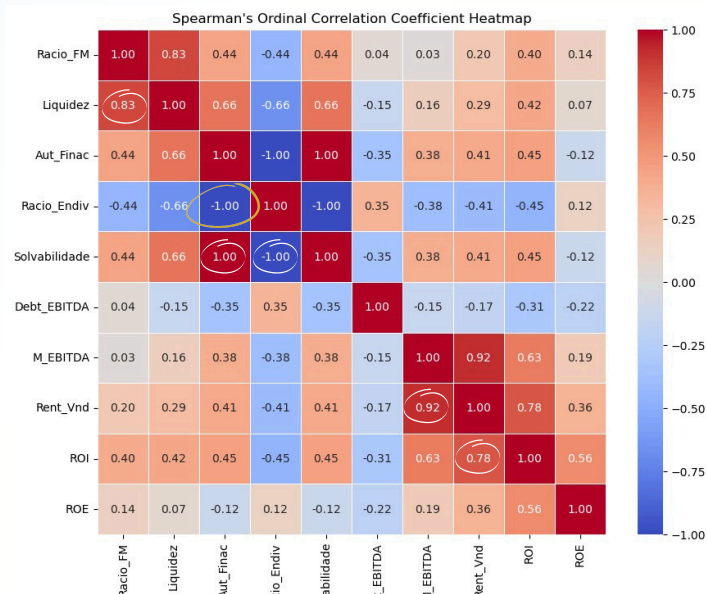


- Large range
- High standard deviation
- Presence of outliers
- Positively skewed distributions

2 groups of companies: “the largest of the largest” (around 5% of the dataset) and “the smallest of the largest” (around 95% of the dataset).

BIVARIATE ANALYSIS

Strong correlation between some financial and economic variables, which remains present in the transformed ratio variables.

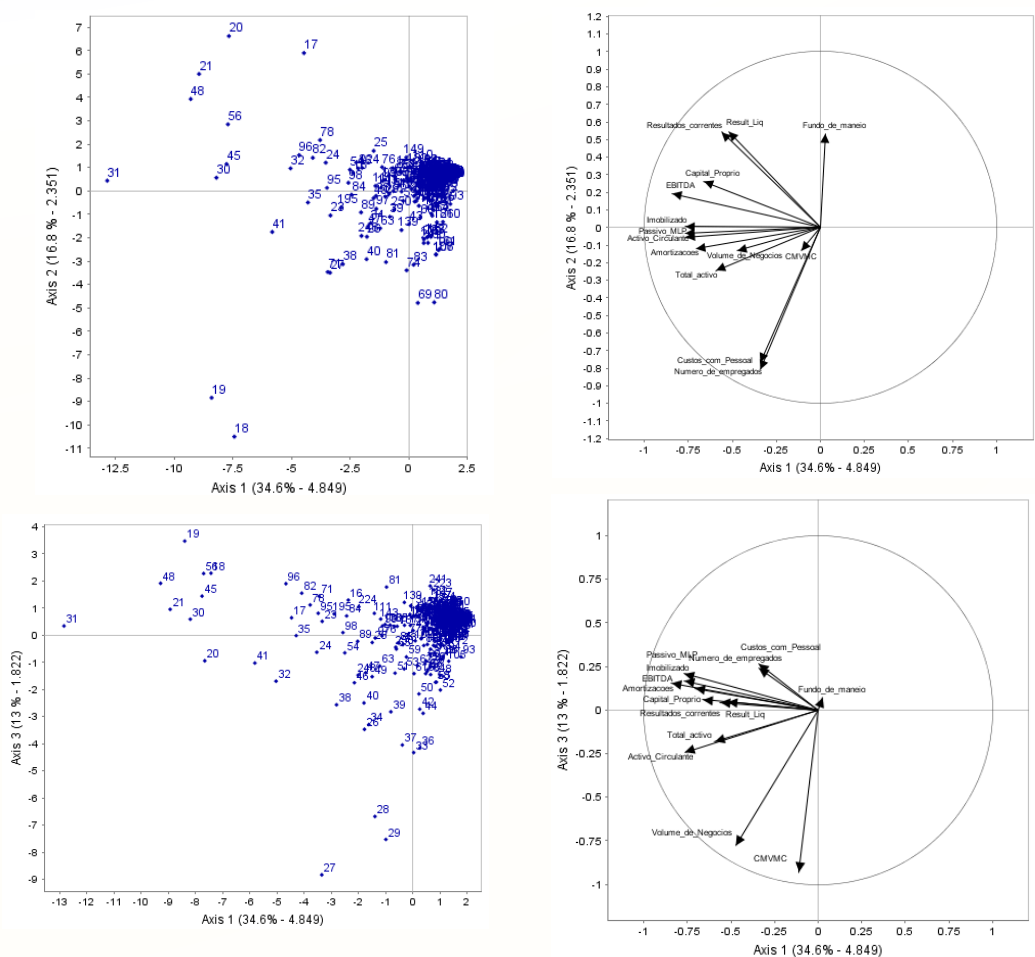


- Kruskal-Wallis test found a statistically significant relationship between CAE_agr (Economic Activity Category) and most financial/economic variables.

PRINCIPAL COMPONENT ANALYSIS

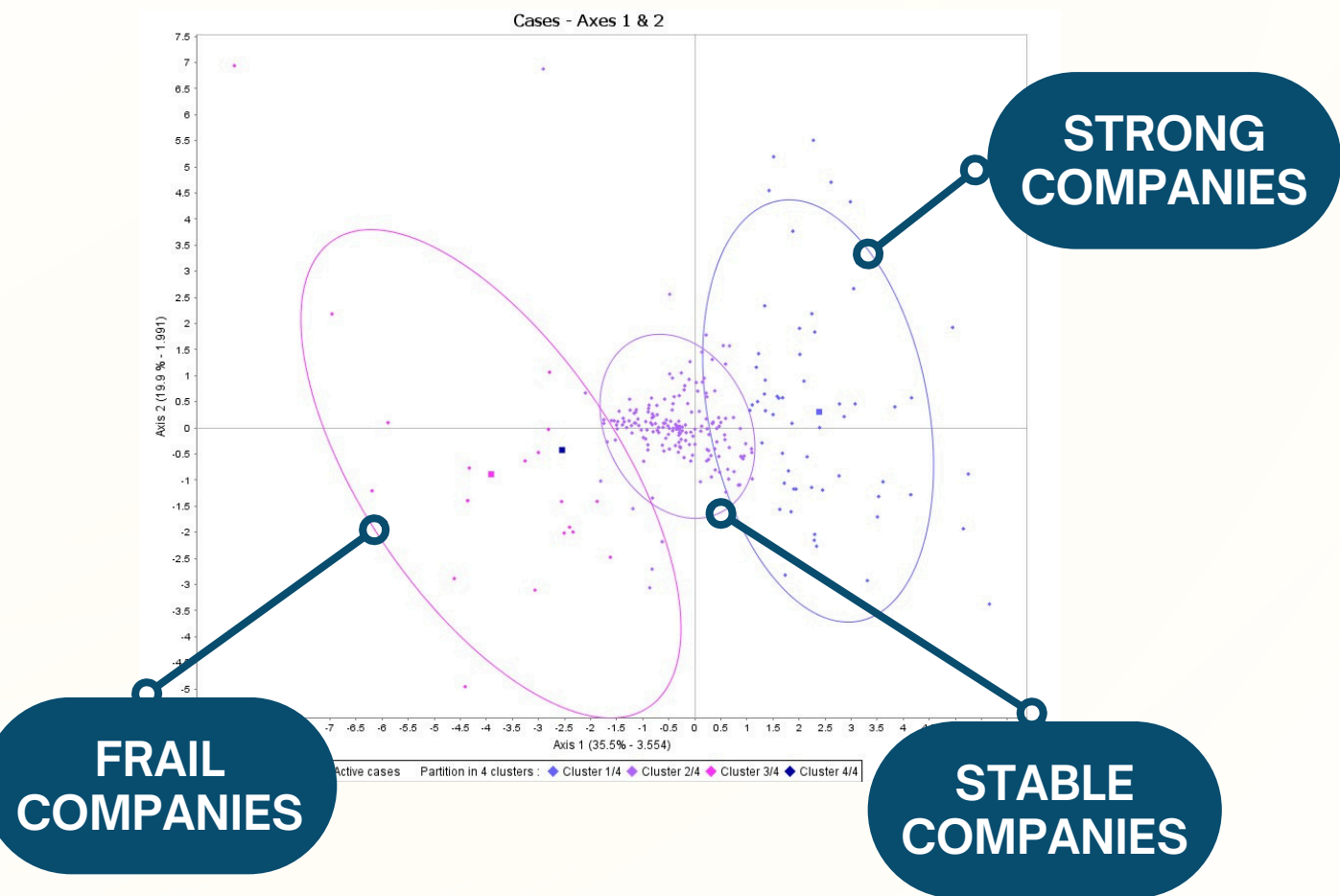
Using Normed PCA and the original 16 variables, we found the main elements through which we can compare and distinguish the companies:

- 1ST AXIS: EBITDA and size of financial structure
- 2ND AXIS: no. of employees and associated costs
- 3RD AXIS: sales margin (sales volume and costs)



CLUSTER ANALYSIS

Using the 10 ratio variables (standardized), Squared Euclidian Distance and Ward's Linkage as the aggregation index, we obtained 3 well defined clusters:



The largest cluster (Stable Companies) is a compact cluster while the others are more dispersed: most companies are in a similar, economically and financially stable situation.

MAIN FINDINGS

- There is, among the top 250, a small group of companies that are **the largest of the largest: around 5% of the dataset** are companies with a turnover between € 2 billion and € 15 billion
- The top 250 is composed of a majority of relatively healthy companies, and two smaller groups of economically and financially poor companies, and of exceptionally good companies. **Around 80% of the companies are financially and economically healthy.**
- The **type of economic activity** (represented by CAE_agr) has a significant effect on the economic and financial structure of the companies.
- The largest portuguese companies can be compared and distinguished through **EBITDA, size of financial structure, number of employees and sales margin.**

Read the full report here:

